

THERESA MARIE LAVERTY

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Department of Fish, Wildlife and Conservation Ecology
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RESEARCH & TEACHING INTERESTS

Vertebrate ecology and conservation, population and community ecology, food web ecology, dryland ecology and management, demographic modeling, bibliometrics

EDUCATION

- Ph.D. in Fish, Wildlife, and Conservation Biology**, Colorado State University 2019
Bats, elephants, and their food: A conservation perspective on trophic interactions in the Namib Desert
Dissertation Advisor: Joel Berger
- B.A. in Ecology and Evolutionary Biology**, Princeton University 2010
Niche overlap in the diets of three Amazonian caiman species
Thesis Advisor: Andrew P. Dobson

PROFESSIONAL APPOINTMENTS

- Assistant Professor**, New Mexico State University 2023-Present
Department of Fish, Wildlife and Conservation Ecology
- Postdoctoral Scholar**, University of Chicago 2021-2022
Department of Ecology and Evolution
- Postdoctoral Scholar**, Colorado State University 2019-2021
Department of Fish, Wildlife and Conservation Ecology

AWARDS & FELLOWSHIPS

AWARDS

- Outstanding Departmental Publication Award, Colorado State University 2021
for the 2021 *Ecological Applications* article

FELLOWSHIPS

- National Science Foundation Graduate Research Fellowship 2013-2018

GRANTS IN SUPPORT OF RESEARCH, TEACHING, & TRAINING

1. New Mexico Department of Wildlife. “Acoustic surveys of bat species of greatest conservation need within wind energy-impacted landscapes of New Mexico” (\$114,066; PI: TM Laverty; 2027-2029).
2. National Science Foundation. “LTER: Advancing ecological understanding of drylands through integrated research at the Jornada Basin (JRN-8)” (\$7,649,997 [3% credit]; PI: NP Hanan, Co-PIs: BT Bestelmeyer, S Bestelmeyer, DM Browning, F Garcia-Pichel, TM Laverty, GE Maurer, ME Mauritz-Tozer, GS Okin, N Pietrasiak, SC Reed, AL Romero-Olivares, OE Sala, RL Schooley, ER Vivoni; 2025-2031).
3. Arizona Game and Fish Department. “Movement ecology of bats with Motus” (\$100,000; PI: TM Laverty; 2025).
4. Bureau of Land Management. “Identifying barriers to pronghorn (*Antilocapra americana*) habitat connectivity in southwestern New Mexico in relation to rangeland management and renewable energy development” (\$502,843 [30% credit]; PI: JW Cain, Co-Pis: JK Frey, TM Laverty; 2024-2028).
5. National Science Foundation. “BRC-BIO: The interactive effects of bats and insects on shrub encroachment dynamics in the Chihuahuan Desert” (\$497,568; PI: TM Laverty, 2024-2027).

6. National Park Service. “Determining activity patterns within and seasonal movements between protected areas in the southwest borderlands for bats of greatest conservation need” (\$54,884; PI: TM Lavery; 2024-2026; with an additional \$42,500 awarded in 2025).
7. Bureau of Land Management. “Influence of Wind Energy Development on Medium and Large Mammals” (\$248,609 [50% credit]; PI: JK Frey, Co-PI: TM Lavery, 2023-2025; with an additional \$5,800 awarded in 2024).
8. National Geographic Society. “Coding for conservation: Quantitative training for the next generation of environmental leaders in Madagascar” (\$25,000; PI: CE Brook, Team Members: AF Andrianiana, E Cornelius Ruhs, TM Lavery, O Razafindramatsima; 2022-2023).
9. New Mexico Department of Game and Fish. “Seasonal abundance, population structure, and diet of long-nosed bats in southwestern New Mexico in relation to contemporaneous food availability” (\$49,839; PI: K Stoner, Co-PI: TM Lavery; 2020-2022).
10. Grants for Graduate Research: National Science Foundation Graduate Research Fellowship, Bat Conservation International Student Research Scholarship for Global Bat Conservation Priorities, National Geographic Society Young Explorers Grant, Sigma Xi Grant-In-Aid of Research, The Explorers Club Exploration Fund, University of Montana (Fred H. Mass Scholarship, Danny On Memorial Scholarship, George and Mildred Cirica Graduate Student Support Fund [declined due to university transfer]), and Colorado State University (Eugene Decker Fellowship, Hill Memorial Fellowship, WCNR Student Success Scholarship) (Total: \$149,450; 2013-2019).
11. Grants for Undergraduate Research: Princeton University (John T. Bonner Senior Thesis Fund and the Dean of the College Fund for undergraduate senior thesis research) (Total: \$7,800; 2009).

PUBLICATIONS & SCHOLARLY COMMUNICATION

Lab personnel: ^{GS}Graduate Student, ^{UG}Undergraduate Student; *Co-First Author

PREPRINTS AND MANUSCRIPTS IN REVIEW or IN REVISION

1. Cooper MC, LaRue ZBT, Boeing WJ, Lavery TM, Fernald AG, Geli HME, Jones KW. Water conservation attitudes and behaviors in the arid southwestern United States. *In review*, **Sustainable Water Resources Management**.
2. ^{GS}Rohan IM, Lavery TM, Frey JK. Terrestrial mammals alter habitat selection in response to turbine visibility at a central New Mexico wind energy facility. *In revision*, **Conservation Science and Practice**.

PEER-REVIEWED PUBLICATIONS & BOOK CHAPTERS

1. Montauban C, Budinski I, Webala PW, Lavery TM, Tanshi I, Torrent L, Bakwo-Fils E, Taylor PJ, Kane A, Monadjem A. 2026. Underrepresentation of bats in Africa’s protected areas. **Conservation Biology** 40(1):e70108. <https://doi.org/10.1111/cobi.70108>
2. Andrianiana AF, Andry S, Kettenburg G, Ranaivoson HC, Lacoste V, Dussart P, Heraud J-M, Lavery TM, Guth S, Young KI, Andrianarimisa A, Brook CE. 2025. Diversity and seasonality of ectoparasite burden on two species of Madagascar fruit bat, *Eidolon dupreanum* and *Rousettus madagascariensis*. **Parasites & Vectors** 18(1):302. <https://doi.org/10.1186/s13071-025-06805-z> **Data & Code:** GitHub
3. Waag AG, Johnson JS, Schorr RA, Frick WF, Lavery TM, Neubaum DJ, Siemers JL, Treanor JJ, Reyes GA, Halstead BJ. 2025. Using passive integrated transponder (PIT) tags to monitor and research bats: applications, benefits, and limitations. **Journal of North American Bat Research** Special Issue 1:100–118. <https://www.eaglehill.us/nabr-pdfs-special/nabr-008-Waag.pdf>
4. Lavery TM, Buecher DC, Dalton DC, Davies ML, Gual-Suárez F, Heady III PA, Medellín RA, Ordóñez-García MC, Pérez-Harp S, Rivera-Villanueva AN, ^{GS}Stevenson BD, Stoner KE, Wolf SA, Zamora-Gutierrez V, Zavala-Norzagaray AA, Frick WF. 2025. Long-haul flights and migratory routes of a nectar-feeding bat. **Ecology** 106(4):e70067. <https://doi.org/10.1002/ecs.70067> **Data & Code:** Zenodo

5. Lavery TM, Buecher DC, Dalton DC, Davies ML, Gual-Suárez F, Heady III PA, Medellín RA, Ordóñez-García MC, Pérez-Harp S, Rivera-Villanueva AN, ^{GS}Stevenson BD, Stoner KE, Wolf SA, Zamora-Gutierrez V, Zavala-Norazaray AA, Frick WF. 2025. On the move: documenting long-distance flights of a nectar-feeding bat. **The Bulletin of the Ecological Society of America** 106(3):e70016. <https://doi.org/10.1002/bes2.70016>
6. Monadjem A, Montauban C, Webala PW, Lavery TM, Bakwo-Fils EM, Torrent L, Tanshi I, Kane A, Rutrough AL, Waldien DL, Taylor PJ. 2024. African bat database: curated data of occurrences, distributions and conservation metrics for sub-Saharan bats. **Scientific Data** 11(1):1309. <https://doi.org/10.1038/s41597-024-04170-7> **Data & Code:** [figshare](#)
7. *Nelson-Olivieri JR, *Layden TJ, Antunez E, Khalighifar A, Lasky M, Lavery TM, Sanchez KA, Shannon G, Starr S, Verahrami AK, Bombaci SP. 2024. Inequalities in noise will affect urban wildlife. **Nature Ecology & Evolution** 8(1):163–174. <https://doi.org/10.1038/s41559-023-02257-9> **Data:** [Dryad](#) **Code:** [Zenodo](#)
8. Wilkins K, Carroll SL, Davis KP, Hauptfeld R, Jones MS, Larson CL, Lavery TM, Pejchar L. 2023. Sexual harassment disproportionately affects ecology and evolution graduate students with multiple marginalized identities in the United States. **BioScience** 73(5):376–387. <https://doi.org/10.1093/biosci/biad032>
9. Lavery TM, Stoner KE. 2022. In search of bachelorettes: observations of male *Leptonycteris yerbabuenae* with dorsal patches across its range. **Therya** 13(2):163–170. <https://doi.org/10.12933/therya-22-1146> **Media:** [figshare](#)
10. Mushabati LM, Eiseb SJ, Benda P, Lavery TM. 2022. Effects of lunar phase and temperature on bat activity and species richness at varying altitudes in the Kunene Region, Namibia. **African Journal of Ecology** 60(3):467–480. <https://doi.org/10.1111/aje.12968>
11. Lavery TM, Berger J. 2022. Indirect effects of African megaherbivore conservation on bat diversity in the world's oldest desert. **Conservation Biology** 36(2):e13780. <https://doi.org/10.1111/cobi.13780> **Data & Code:** [figshare](#)
12. Lavery TM, Teel TL, Gawusab AA, Berger J. 2021. Listening to bats: Namibian pastoralists' perspectives, stories, and experiences. **Journal of Ethnobiology** 41(1):70–86. <https://doi.org/10.2993/0278-0771-41.1.70>
13. Mans BJ, Kelava S, Pienaar R, Featherston J, de Castro MH, Quetglas J, Reeves WK, Durden LA, Miller MM, Lavery TM, Shao R, Takano A, Kawabata H, Moustafa MAM, Nakao R, Matsuno K, Greay TL, Evasco KL, Barker D, Barker SC.. 2021. Nuclear (18S-28S rRNA) and mitochondrial genome markers of *Carios (Carios) vespertilionis* (Argasidae) support *Carios* Latreille, 1796 as a lineage embedded in the Ornithodorinae: re-classification of the *Carios* sensu Klompen and Oliver (1993) clade into its respective subgenera. **Ticks and Tick-borne Diseases** 12(4):101688. <https://doi.org/10.1016/j.ttbdis.2021.101688>
14. *Aubry LM, *Lavery TM, *Ma Z. 2021. Impacts of COVID-19 on ecology and evolutionary biology faculty in the United States. **Ecological Applications** 31(2):e2265. <https://doi.org/10.1002/eap.2265>
15. Orlova MV, Lavery TM, Reeves WK, ^{UG}Gratton EM, Davies ML, Anisimov NV. 2020. New geographical and host record of bat ectoparasite *Steatonyssus (Steatonyssus) afer* Radovsky and Yunker, 1963 (Mesostigmata: Gamasina: Macronyssidae). **Acarologia** 60(4):951–958. <https://doi.org/10.24349/acarologia/20204411>
16. Reeves WK, Lavery TM, ^{UG}Gratton EM, Mushabati LM, Eiseb SJ. 2020. New national records for *Cyclopodia greeffi greeffi* (Diptera: Nycteribiidae) from the Kunene Region, Namibia, Africa. **Entomological News** 129(3):327–329. <https://doi.org/10.3157/021.129.0311>
17. Reeves WK, Mans BJ, Durden LA, Miller MM, ^{UG}Gratton EM, Lavery TM. 2020. *Rickettsia hoogstraalii* and a *Rickettsiella* from the Bat Tick *Argas transgaripepinus*, in Namibia. **Journal of Parasitology** 106(5):663–669. <https://doi.org/10.1645/20-46>
18. Lavery TM, Berger J. 2020. Do bats seek clean water? A perspective on biodiversity from the Namib Desert. **Biological Conservation** 248:108686. <https://doi.org/10.1016/j.biocon.2020.108686> **Data & Code:** [Mendeley Data](#)
19. Lavery TM. 2020. The Namib Desert and its bats. In: Goldstein MI, DellaSala DA, editors. *Encyclopedia of the world's biomes*, vol. 2. Oxford: Elsevier; p. 181–192. <https://doi.org/10.1016/B978-0-12-409548-9.12011-1>

20. Orlova MV, Lavery TM, Reeves WK, ^{UG}Gratton EM, Davies ML. 2020. The first finding of parasitic mite, *Parasteatonyssus nyctinomi* (Mesostigmata: Gamasina: Macronyssidae), in Namibia. **Journal of the South African Veterinary Association** 91(1):a2002. <https://doi.org/10.4102/jsava.v91i0.2002>
21. Orlova MV, Lavery TM, Reeves WK, ^{UG}Gratton EM, Davies ML. 2020. The first record of the spinturnicid mite *Spinturnix kolenatii* Oudemans, 1910 (Mesostigmata: Gamasina: Spinturnicidae) from the long-tailed serotine bat *Eptesicus hottentotus* A. Smith, 1833 (Chiroptera: Vespertilionidae) in Africa. **International Journal of Acarology** 46(3):160–164. <https://doi.org/10.1080/01647954.2020.1731596>
22. *Jimenez MF, *Lavery TM, Bombaci SP, Wilkins K, Bennett DE, Pejchar L. 2019. Underrepresented faculty play a disproportionate role in advancing diversity and inclusion. **Nature Ecology & Evolution** 3(7):1030–1033. <https://doi.org/10.1038/s41559-019-0911-5>
23. Lavery TM, Teel TL, Thomas REW, Gawusab AA, Berger J. 2019. Using pastoral ideology to understand human-wildlife coexistence in arid agricultural landscapes. **Conservation Science and Practice** 1(5):e35. <https://doi.org/10.1111/csp2.35>
24. Moldowan PD, Lavery TM, Emmans CJ, Stanley RC. 2016. Diet, gastric parasitism, and injuries of caimans (*Caiman*, *Melanosuchus*, and *Paleosuchus*) in the Peruvian Amazon. **South American Journal of Herpetology** 11(3):176–182. <https://doi.org/10.2994/SAJH-D-16-00012.1>
25. Lavery TM, Dobson AP. 2013. Dietary overlap between black caimans and spectacled caimans in the Peruvian Amazon. **Herpetologica** 69(1):91–101. <https://doi.org/10.1655/HERPETOLOGICA-D-12-00031>

COURSE DEVELOPMENT & TEACHING EXPERIENCE

Undergraduate Courses, New Mexico State University

FWCE 330: Natural History of the Vertebrates (2024-2026)

FWCE 355: Wildlife Management and Analysis (2025)

FWCE 431: Mammalogy (2024, 2026)

Graduate Courses, New Mexico State University

FWCE 531: Mammalogy (2024, 2026)

Intro to GitHub Workshop (2023)

Guest Lectures & Participation in Field Courses

New Mexico State University: FWCE 464: Management of Aquatic and Terrestrial Ecosystems (2026), FWCE 330: Natural History of the Vertebrates (2023)

University of British Columbia, Okanagan: BIOL_O 370: African Savannah Biology (2025)

Colorado State University: NR 300: Biological Diversity (2018, 2019, 2020), BZ 330: Mammalogy (2020), FW 455: Principles of Conservation Biology (2020), FW 370: Design of Fish and Wildlife Projects (2018), FW 104: Wildlife Ecology and Conservation (2017)

University of Montana: WILD 480: The Upshot: Applied Wildlife Management (2014, 2015), BIOO 101N: Montana Wildlife (2013)

Princeton University: EEB 380: Ecology and Conservation of African Landscapes field course (2012)

Teaching Assistant

Colorado State University: NR 300: Biological Diversity (2018, 2019), FW 370: Design of Fish and Wildlife Projects (2018)

University of Montana: WILD 480: The Upshot: Applied Wildlife Management (2014), BIOO 101N: Montana Wildlife (2013)

Princeton University: EEB 380: Ecology and Conservation of African Landscapes field course (2012)

STUDENT MENTORSHIP

GRADUATE STUDENTS ADVISED

Flor Calderón	NMSU	Wildlife and Fisheries Ecology	Ph.D.	In progress
Ryan Moore	NMSU	Wildlife and Fisheries Ecology	Ph.D.	In progress
Melanie Solis Alvarez	NMSU	Fish, Wildlife and Conservation Ecology	M.S.	In progress
<i>(Co-advisor: Wiebke Boeing)</i>				
Matthew Becker	NMSU	Fish, Wildlife and Conservation Ecology	M.S.	In progress
<i>(Co-advisor: Jennifer Frey)</i>				
Taylor Pichler	NMSU	Fish, Wildlife and Conservation Ecology	M.S.	In progress
Brandi Stevenson	NMSU	Fish, Wildlife and Conservation Ecology	M.S.	In progress
Holly Whited	NMSU	Fish, Wildlife and Conservation Ecology	M.S.	2026
Iona Rohan	NMSU	Fish, Wildlife and Conservation Ecology	M.S.	2026
<i>(Co-advisor: Jennifer Frey)</i>				

Students under my supervision have been awarded over \$12,000 in grants and awards in support of their research, teaching, and outreach and been the recipients of the following fellowships- and grants to support their studies:

- JRN-LTER Graduate Research Fellowship (R Moore)
- Ecological Society of America's Forrest Shreve Student Research Award (T Pichler)
- Southwestern Association of Naturalists' Howard McCarley Student Research Grant (H Whited)

DOCTORAL COMMITTEE SERVICE

Ryan Rimple	NMSU	Biology	Ph.D.	In progress
Mark Peyton	NMSU	Biology	Ph.D.	In progress

MASTERS COMMITTEE SERVICE

Nick Sebastian	UIUC	Natural Resources and Environmental Sciences	M.S.	In progress
Charisa Bell	NMSU	Fish, Wildlife and Conservation Ecology	M.S.	In progress
Julia Nelson	NMSU	Fish, Wildlife and Conservation Ecology	M.S.	2026
William Grooms	NMSU	Fish, Wildlife and Conservation Ecology	M.S.	2025
Elizabeth Brock	NMSU	Water Science and Management	M.S.	2025
Samuel Martinez	NMSU	Fish, Wildlife and Conservation Ecology	M.S.	2024

EXTERNAL THESIS EXAMINER

Veli Monday Mdluli University of the Free State QwaQwa (S. Africa) Ph.D. Thesis Examiner 2024

SERVICE TO THE UNIVERSITY & PROFESSIONAL ORGANIZATIONS**NEW MEXICO STATE UNIVERSITY**

Institutional Animal Care and Use Committee (2023-present)

NMSU COLLEGE OF AGRICULTURAL, CONSUMER AND ENVIRONMENTAL SCIENCES

Connections & Engagement Initiative Committee (2025-present)

NMSU DEPARTMENT OF FISH, WILDLIFE AND CONSERVATION ECOLOGY

Undergraduate Curriculum Committee (2024-present)

Search Committee, Chair of the Department of Fish, Wildlife and Conservation Ecology (2024)

Diversity, Equity, and Inclusion Committee (2023-2024)

JORNADA BASIN LTER

Information Management Committee (2025-present)

OTHER UNIVERSITIES & PROFESSIONAL ORGANIZATIONS

State Wildlife Action Plan (SWAP) for New Mexico Core Team, New Mexico Department of Wildlife (2025-present)
Co-coordinator of Southern African bat assessments for the IUCN Red List (2023-present)
GBatNet Prioritizing Key Habitats and Species Conservation Working Group (2022-present)
NABat Emergence/Summer Colony Count Working Group (2020-2024)
Lesser Long-nosed Bat Post-Delisting Monitoring Plan Participant, U.S. Fish and Wildlife Service (2023)
Diversity, Equity, & Inclusion Committee, Ecology and Evolution Department, University of Chicago (2021-2022)
Planning Committee Member, Northern Colorado & Wyoming chapter of Expanding Your Horizons—an annual STEM workshop for 6th through 8th grade girls (2017-2019)
Dr. David Otis Speaker Series Selection Committee, Colorado State University (2017-2018)
Panel member, Colorado State University Info Session on the NSF Graduate Research Fellowship Program (2017)
Graduate Representative, Population Ecology Assistant Professor Search Committee, University of Montana (2013-2014)
Graduate Representative, Department of Ecosystem & Conservation Sciences Student Evaluation Committee, University of Montana (2014)

SERVICE AS A REVIEWER

Ad hoc journal referee (No. of reviews): *Acta Chiropterologica* (5), *Biological Conservation* (2), *Biotropica* (2), *EcoHealth* (1), *Global Change Biology* (1), *Global Ecology and Biogeography* (3), *Herpetological Monographs* (1), *Human Dimensions of Wildlife* (7), *Journal of Bat Research and Conservation* (1), *Journal of Mammalogy* (5), *Mammal Review* (1), *PLOS ONE* (1), *Proceedings of the Royal Society B: Biological Sciences* (2), *Scientific Reports* (1), *Sustainability* (2), *Urban Ecosystems* (2)

Other reviewer service: Bat Conservation International (8), National Geographic Society (17), The Red List of Mammals of South Africa, Lesotho and Swaziland (1, species status assessment)

SEMINARS, OUTREACH, & MEDIA APPEARANCES

INVITED SEMINARS

2026 Angelo State University, Dept. of Biology
University of Nevada, Las Vegas, School of Life Sciences
National Park Service Natural Resources Stewardship and Science Directorate webinar
2024 Jornada Basin LTER Desert Ecology Short-Course
2023 Iowa State University, Dept. of Natural Resource Ecology and Management
North American Bat Monitoring Program (NABat) Community of Practice call
New Mexico State University, Dept. of Biology
2022 University of Illinois Urbana-Champaign, Program in Ecology, Evolution, and Conservation Biology
New Mexico State University, Dept. of Fish, Wildlife and Conservation Ecology
2021 Bat Conservation International, Brown Bat Lunch series
Rutgers University–New Brunswick, Graduate Program in Ecology and Evolution
Society for Conservation Biology, Emerging Issues in Conservation series
Indiana State University, Dept. of Biology
University of Northern Colorado, Dept. of Biology
2020 Boise State University, Dept. of Biological Sciences
Oak Ridge National Laboratory, Aquatic Ecology Group

INTERVIEWS & MEDIA APPEARANCES

2025 KRQE News 13-Albuquerque
2021 Roan NEWS Magazine, Namibian Environment & Wildlife Society
2020 Nature Research Ecology & Evolution Community
2019 Nature Research Ecology & Evolution Community
2018 Rocky Mountain Public Broadcasting Station, The Conservation
2016 National Geographic Society

PROFESSIONAL DEVELOPMENT

TRAINING PROGRAMS

- 2013 University of Florida Science Communication Academy
- 2004 California Academy of Sciences: The Ant Course: Taxonomy & Systematics (Costa Rica)
- 1996 Organization for Tropical Studies: Tropical Plant Systematics 96-9 (Costa Rica)
- 1993 University of California Education Abroad Program: Tropical Biology Quarter (Costa Rica)

MEMBERSHIPS IN PROFESSIONAL SOCIETIES

North American Society for Bat Research, Society for Conservation Biology, American Society of Mammalogists, Ecological Society of America, Association for Tropical Biology and Conservation, Global South Bats, Western Bat Working Group, The Wildlife Society

LANGUAGES

Fluent in English & Spanish